

41068 ROBOTICS STUDIO 1

Aero-Aid: Remote First Aid Delivery System

Component 1: Proposal Document

TEAM DETAILS

Group number: Team 4

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1. Team and Collaboration

1.1 Team Capabilities and Learning Goals

Team member	Relevant experience or existing skills	Skills they want to develop	Intended areas of contribution
Harry	• Computer vision basics (Lidar, Sonar) • Environment Development (Blender) • Communication	• In depth computer vision and analytics • Map generation through computer vision	• Perception and sensor integration • World generation and user environment
Jacob	• CAD (Solidworks, Onshape, Fusion360) • FEA (Simscale) • Programming (Python, C++, Visual Basic, C#, Java) • Sensor-microcontroller interfacing and data parsing (LiDAR, Microwave Radar, Ultrasonic, IR ranger)	• Sensor fusion • Environment mapping • Advanced terrain navigation	• Graphical User Interface (GUI) • Task planning and co-ordination
Jayden	• CAD development (Solidworks) • Presentation Skills (PowerPoint, Canva) • AI Development (Claude Code) •	• Environment Mapping	• Sensor integration and perception
John	• Programming (C++ & Python) • CAD development • Presentation experience (Figma, Canva, PowerPoint) • Project Management • Medical Science • CAD development (SolidWorks)	• Simulation environments • Path Planning	• Path planning and navigation logic for both drones, • Support simulation environment setup and stakeholder communication.
Will	• Project Management • CAD (Solidworks) • Programming (Python & C++) • Communication	• Computer vision • Path planning	• Path planning • Multi-drone co-ordination

1.2 Team Operating Agreement

In order to establish a productive working relationship between group members, the team created a expectations table that defines each team members responsibility.

Section	Expectation
Attendance and weekly contribution	Each team member is expected to at a minimum attend the weekly tutorial and contribute meaningfully in this tutorial section. Depending on the week and workload, other team meetings may be organised and all team members should attend if available. The team aims to make meaningful contributions each week to ensure workload doesn't overflow towards the end of the project.
Communication outside of class	A teams group chat will be created as a primary form of communication. Instagram will be used as a secondary out of hours contact.
Response to teams messages	For teams chats sent within working hours the expected response time is by end-of-day considering some team members have other jobs and commitments. Out of hours communications on Instagram should be responded to within 24 hours.
Decisions made and recorded	All major design decisions that impact all team members will be made by group discussions and if necessary, group vote. This will be recorded in communication channels. All minor design decisions specific to each role can be decided by the relevant team member.
Responsibility and deadline management	The team leader will organise high level deadlines, Gant chart and deliverables which will be broken down into smaller sub-tasks by relevant team members. Deliverable deadlines will be organised with redundancy.
Establishing Uneven Contribution & Disagreements	Any uneven contributions or disagreements should be escalated and addressed by the team leader. If an agreement is not reached through mediation with the team leader and the rest of the group, the issue should be escalated to teaching staff.
Standard of work	All team members are expected to produce quality, authentic work and contribute meaningfully to all group discussions.

2. Project Direction

2.1 Candidate Applications

Candidate application	Intended user or stakeholder	Need or potential value	Main technical challenge	Feasibility this semester
Remote First Aid Delivery System	Rescue teams such as SES, Rural fire service or park rangers	Providing expedited first aid equipment delivery to people stranded in remote areas that are unable to be reached easily on foot.	Communication between both drones to establish the terrain and a path to get to the human	Highly feasible with 5-person team to co-ordinate the drones to establish surveying, path planning and actuation
Biodiversity credit scheme identifier	Either used by ecologists to aid in environmental surveys or by landowners to continuously	The NSW biodiversity credits scheme requires ecologists to conduct continuous surveys to understand the value of the land.	Drones ability to detect environmental features and geography to the standard required by the NSW	Difficult to produce a product that exactly aligns with the scheme but still feasible to provide an aid

	assess land value	Continuous drone surveyance would expedite the audit processes involved in this scheme	government for the scheme.	tool that assists in this process.
Illegal dumping detector	Governments or park rangers responsible for nature preservation	Quickly survey large areas for illegal dumping, littering, etc. to protect nature areas and respond quickly to illegal dumping	Identifying what illegal dumping looks like given it had various appearances and sizes	Medium feasibility, only difficulty is perception and identification of illegal dumping.
Forrest fire and tree health detection	Park rangers or the rural fire service	Survey rural areas for fire risks or active fires to address early	Using thermal imaging on the drones	Much simpler project that is very feasible however lacks technical depth compared to other projects.
Land stewardship mapping for rangers	Indigenous land ranger / land management organisation	Rangers caring for country need an accessible way to track vegetation and terrain changes over time to prioritise where stewardship effort is most needed.	Reliable map alignment and change detection across repeat missions with noisy sensor data.	Feasible with a 5-person team. Limited by short window for repeat surveys.

2.2 Chosen Application and Stakeholder Need

The system we propose is intended to be used primarily by search and rescue co-ordinators, park rangers and medical response teams. This is specifically for the use case of responding to lost or injured personnel in bushland, forest or national parks where vehicle access is slow or impossible. When people experience significant injuries in remote areas, it is critical to provide immediate first aid as this significantly increases the patients survival chances. Implementing a robotic system that uses aerial drones to cross difficult terrain to a person before responders physically reach the sight can aid the patient with valuable medical supplies and communication to paramedics without risking human searchers on hazardous terrain. This system would aid in both finding missing people in these remote areas and then providing them with medical supplies once found.

This application was selected over other candidate ideas such as vegetation surveying or environmental modelling concepts because it encapsulated a more challenging task across perception, path planning and delivery. This task application supersedes our other ideas because of its unique use of the required forest environment. The project is feasible to be completed within the subject timeline because the scope is still bounded by two well defined drones with clean interfaces pre-built.

2.3 Related Work

Existing research, product or system	What it does	Relevant strengths or limitations	How it informs your project
Everdrone - Emergency Medical Aerial Delivery (EMADE) Blecher, D. (2022)	The EMADE is a Swedish built drone which was the first ever emergency drone to save a patient's life from a serious cardiac event.	It's size allows for large packages to be flown, including AED's. It's size also is a limitation for navigation in tighter, more tree-dense areas.	For our application, a large drone like this is necessary for the package, and shows that it is possible in the real world. However, it shows the challenge of manoeuvrability.
Fire and Rescue NSW - Remote Pilot Aircraft System (RPAS) ("Vehicles - Fire and Rescue NSW," 2026) (Fire and Rescue NSW Drone Team Locates Missing Hikers Using AI Technology, 2026)	FRNSW mainly use their RPAS system for search and rescue. It is equipped with both an IR and coloured camera and can be remotely controlled by a pilot.	Having multiple cameras allows for the drone to be used in a variety of situations. Especially using an IR camera allows for night searches. However, as it is small it can only be used for search and not to help people once they are found.	Looking at the strengths of this drone, it makes sense to have a smaller drone for rapid search. It is also less resource intensive and would be able to search for longer.
TU Delft Ambulance Drone	Provides rapid response to cardiac arrest scenarios via a defibrillator-equipped drone capable of being deployed in dense, urban environments.	Due to size and weight constraints enforced by the drone's payload capacity, any additional equipment cannot be equipped. Due to its compact size and specialised nature, it is highly effective in delivering a defibrillator within minutes.	Provides a real-world project example of a life-saving drone system akin the one our group intends on developing, albeit at a more sophisticated level.

2.4 Long-Term Concept of Operations

The current proposed system has application to many different use-cases on a much broader scale. A full-scale application of the first-aid response system would have multiple remote and portable systems that could be utilised across the state to have full-coverage response to remote emergencies. This would include designated launch spots at popular hiking trails and nature areas but also portable launch stations operated by SES vehicles to support search and rescue in any location, operated from the vehicle. This would create a network of drones across areas of interest working as one cohesive system.

In addition to this, popular camping sites, nature trails or other high risk areas, small location distress beacons would be given to hikers. In case of emergency, a distress signal would automatically trigger a response from the first-aid response system autonomously without human assistance.

2.5 Concept of Operations – For 41068 Robotics Studio 1

The primary goal for this subject is to create a first aid response system in remote forest environment that can find a stranded person and deliver a first aid kit to them. The system operates on a two drone architecture, in which one drone acts as the scout and the second serves as the courier, delivering first aid supplies once a location is confirmed. Through the UI, a user will set a designated area on a map to conduct the search. Upon confirmation of the search grid, the scout drone methodically scans the search grid looking for the stranded person. Once the stranded person is found, the scout reports back to base photos and co-ordinate for the user to confirm whether the drones detection is accurate. Once confirmed by the user, the scout drone stays and monitors the situation and the 2nd courier drone launches. The courier drone must find the fastest path to the co-ordinates provided and deliver the first aid kit. Throughout the operation, the user receives constant feedback on each drones status and can manually abort the mission based off information provided by the drones.

2.6 Scope Boundaries and Assumptions

Category	Current definition
The system will...	- Plan a logical path to fly over given set search parameters (search grid), locate and identify stranded people, provide consistent live feedback to the drone operator via user interface, deliver a first aid kit to the identified stranded people via a second drone, have interactive GUI features such as emergency stop.
The system will not...	- Diagnose injuries, give medical advice, make completely autonomous decisions, fly beyond the mapped environment, replace human response systems.
Environmental assumptions	- The search area is bound, general forest environment, the drone will “drop” a first aid kit from a fixed Z value, no collisions with leaves,
Robot or sensor assumptions	- Both drones have accurate drift-free position estimations, RGB and thermal camera have fixed known calibrations, courier drone carries first aid kit with no affect on drone physics
User or operator assumptions	- The user knows how to operate the GUI, the user is familiar with the search area, the user understands technical feedback from the drone
Simulation simplifications	- Trees have no collision detection with trees, humans have set heat values with no radiative properties, drones have no accurate physics such as gravity,
Other important constraints	The MVP product will be designed on a single drone system then expanded into a 2-drone system on the stretch scenario.

2.7 Project Title

Aero-Aid

Aero-Aid is a 2-drone search and response system that can accurately and expediently locate stranded individuals in forest environments and deliver first aid equipment.

3. System Design

3.1 System Requirements

- **Perception:** how the system senses, detects, maps, estimates, classifies, or otherwise interprets relevant information;
- **Decision making:** how the system selects actions, plans motion, responds to information, allocates tasks, or makes mission-level choices;
- **User interface:** how useful information, status, alerts, maps, mission progress, or controls are provided to a user or operator.

Divide the requirements into MVP requirements that must be working by the final demonstration and stretch requirements for advanced functionality.

Some examples have been provided to get you started – you are welcome to edit these or remove them.

ID	Category	Requirement	Why is it important?	How will it be verified or demonstrated?	Priority: MVP or Stretch
R1	User Interface	The user can set the search grid for the drone to search	Used for path planning and essential for project modularity	The decision-making system receives accurate relevant data that represents the search grid	MVP
R2	Decision Making	The system can plan a logical search grid path given a search grid and actuate it	Core backbone behind the person detection system.	Visual output feedback of drones planned path based on search grid	MVP
R3	User Interface	The user receives live visualised and quantitative feedback of drone position and detection	Useful for operator to have and make actions based off	User interface is easy to use, logical, and displays all relevant or helpful information for the operator	MVP
R4	Perception	People are detected via RGB camera using computer vision	Key to demonstrating human detection and providing aid	People inside the environment are detected accurately using the camera	MVP
R5	User interface	Once detected, operator receives RGB image to confirm human detection and initiate first aid response	Minimises computer vision hallucinations by adding human-interface framework	GUI displays camera image when human is detected where user can confirm/deny the presence of a human	MVP Stretch?
R6	Decision Making	The drone plans and actuates a path from its current location, to the first aid kit and back to the user	Essential to delivering the first aid and not just finding the person	Visual confirmation in simulation that drone can plan and actuate the given path	MVP
R7	Perception	The drone can detect any potential collisions that the first aid kit would hit upon delivery and adjust the drop site	In practice, the drone must find an appropriate drop-site to ensure the aid kit actually reaches the target person.	Simulation test with obstructed drop site.	Stretch

R8	Decision Making	A second courier drone is used to deliver first aid to the user	In practicality, a second drone would provide quicker response times and would allow for a larger less manoeuvrable second drone well equipped for carrying a large first aid kit	Simulation shows second drone taking off and delivering first aid to user increasing overall response time	Stretch
R9	Decision Making	Data is transferred from the scout drone to the courier drone for location positions	Allows for second drone path planning	Data transfer feedback seen in terminal	Stretch
R10	Perception	People are detected/confirmed via thermal imaging camera	Ensures increased accuracy in people detection	People detection has increased accuracy based on thermal and RGB detection	Stretch
R11	Simulation	Simulation has randomly generated terrain including random location of stranded human	Allows for easy simulation of multiple environment cases without having to change code	Upon running each test, the environment is noticeably different	Stretch
R12	Simulation	Simulation has a night time random event to depict night searches.	There are many scenarios in real life where people go missing at night, and a drone must be able to function at these times.	The environment will have reduced light and it will be darker.	Stretch
R13	Decision Making	Scout drone can track moving human while courier drone is reaching location	Allows the injured person to move and be monitored without the drone losing its location	Simulation with moving human still delivers first aid	Stretch

3.2 System Diagram

The figure below is the high-level overview of the system functionality, where the GUI will serve as the mission control and manage and distribute all tasks to the 2 sub-drones. This ensures that each drone has all its functionality stored within their own namespaces and these tools are called by the mission control. This architecture also complements the human confirmation design of the system, allowing the operator to have full control at all times and confirm drone decisions through each step.

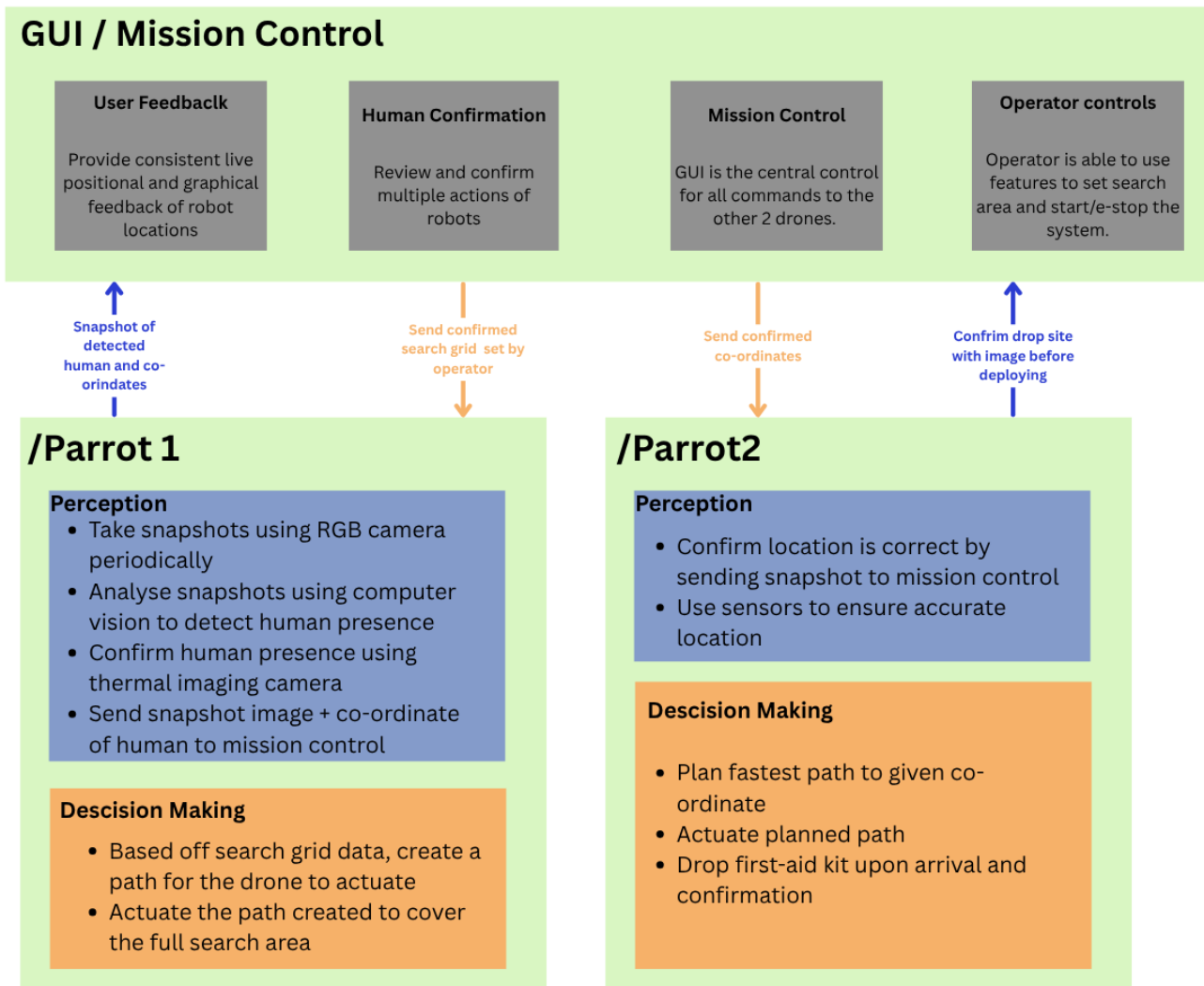


Figure 1- High-level overview of the system functionality

The figure below demonstrates a more technical flow chart of the ROS application of the project

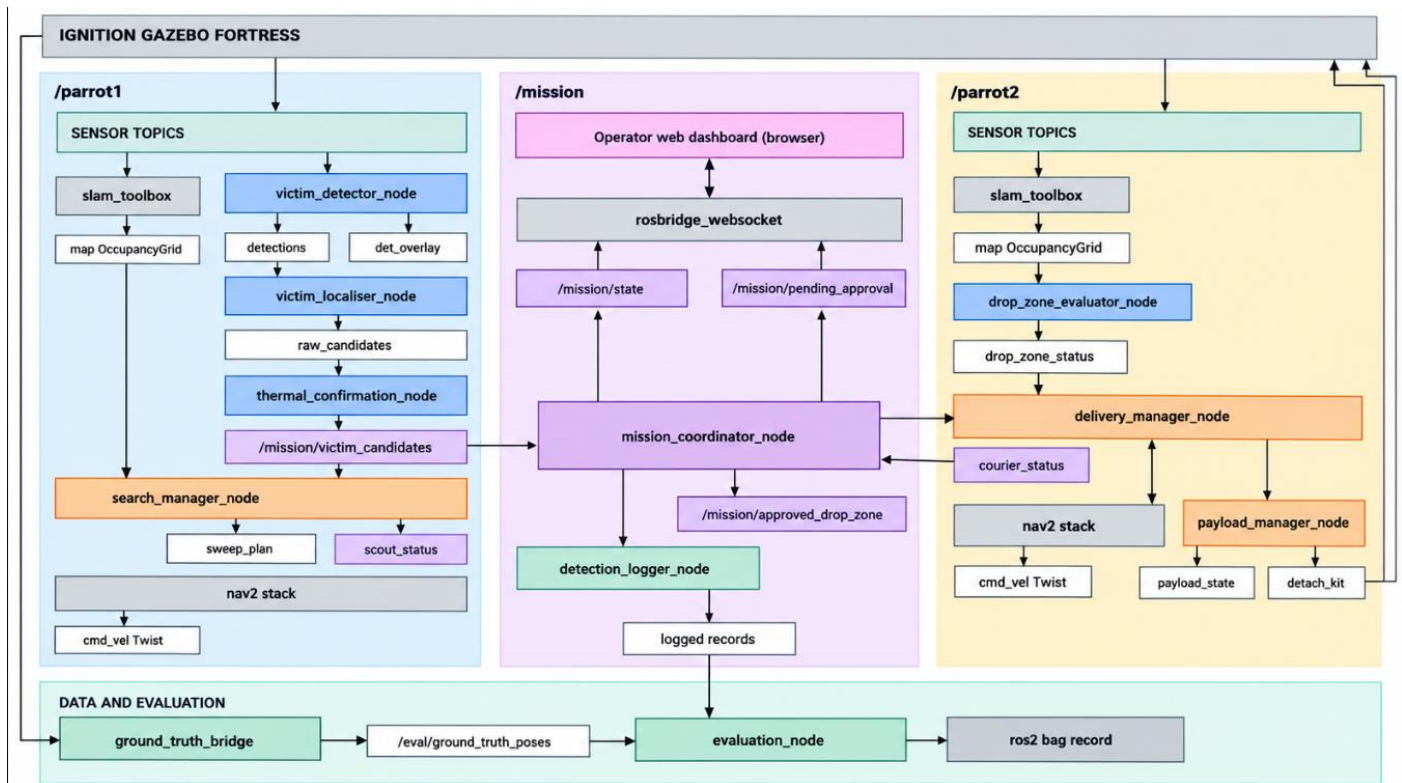


Figure 2- Flowchart

3.3 Component Plan

Component	Purpose	Inputs	Outputs	Provided, existing, adapted, or custom?	Initial lead and support team members
Simulation environment (world)	Forest environment the robots search and operate within	Randomised generation parameters	The physical environment all sensors perceive and robots navigate through	Provided , adapted (base simple_trees/large_demo worlds came with the package; team is adding a person model, search-grid-relevant features, and denser/harsher elements such as rock cliffs)	Harry
Drone platform/robot model	The simulated scout and courier drones themselves, including their sensor suite.	Cmd_vel velocity commands	Simulated sensor data (camera, IMU, lidar) and odometry	Provided (base Parrot URDF/xacro + sensors came with the package); adapted where the team adds the thermal camera sensor (stretch)	Jayden & Harry (lead)

Mapping	Builds a map of the search area as the drone explores	Distance-sensor readings, drone motion	A map of what's been explored	Existing algorithm (slam_toolbox), provided config	John & Will (lead)
Path execution	Drives the drone to a chosen location, avoiding obstacles	Map, target location	A target direction/speed for the drone	Existing algorithm (Nav2), provided config	John & Will (lead)
Localisation	Estimates the drone's current position	Motion and orientation sensor data	A stable position estimate	Existing algorithm (robot_localization), provided config	John & Will (lead)
Simulation-software bridge	Connects the simulated drone's sensors/motors to the rest of the software		—	Existing tool (ros_ign_bridge), provided config for these robots' sensors	Jacob (lead), Harry (support)
Person detection	Detects a person in the camera view and estimates their location	A live camera image	A detected person's location + confidence	Custom (may use an existing library/model as a sub-part)	Jayden & Harry (lead)
Search planner	Decides where the drone should search next	Current map, drone position	The next place to search	Custom - replaces the provided demo script's random-walk logic	John & Will (lead)
Mission coordinator	Tracks overall mission progress (searching, found, confirmed, delivering)	Detection results, navigation results, operator confirmation	The current mission stage	Custom	Jacob (lead), Will & John (support)
Cross-drone location handoff	Transfers a confirmed detection location from the scout drone's map frame into a navigation goal the courier drone can actually reach.	Confirmed detection location (scout's map frame)	A NavigateToPose goal in the courier drone's frame	Custom	John & Will (lead), Jacob (support)

Delivery control	Moves the drone to the person and releases the first aid kit	A confirmed delivery location	Drone descent + kit released	Custom	John & Will (lead)
Mission dashboard	Let's the operator set the search area, watch progress, confirm detections, see delivery status	Map, detections, drone status	Search-area input, confirmation, abort command	Custom	Jacob (Lead)
Thermal Sensor	Second signal detection signal for higher confidence	Simulated thermal-style image	A detected person's location	Custom	Jayden & Harry (lead)
	(add more rows as necessary)				

3.4 Simulation and Evaluation Environment

Our simulation environment will be quite similar to the environment provided in the package. This is because our intended environment will take place in a national park with several trees, rocks and paths. Furthermore, we also intend to implement the parrot drone which was used in the original package simulation as our scout drone.

However, taking these commonalities into consideration, we also are going to branch out from the original environment to better simulate our chosen scenario. For example, a key benefit to using a remote access drone is the ability to reach hard to access areas. As such we intend to harsher environmental biomes such as rock cliffs, more dense tree areas, etc. to better depict the value of our system compared to simple search and rescue. (Fire and Rescue NSW Drone Team Locates Missing Hikers Using AI Technology, 2026).

As noted as a stretch in section 3.1, we also intend to create a night scenario for the drone. Often in real applications park visitors are only noted as missing outside of daylight hours. As such, it is important to develop our simulation to extend its capabilities and allow it to operative during the night and with low visibility. This will greatly improve the validity in the evaluation of our design improving its use case.

3.5 Technical Risks and Early Tests

Risk or uncertainty	Why it matters	Early test or evidence needed	Mitigation or fallback plan	Owner
Person detection reliability	Project requirements requires camera-based person detection.	Place a visually distinctive person placeholder in a test world early and confirm a first-pass detection method can reliably find it before investing further	Use a simple, distinctive marker rather than requiring a realistic person recognition	Jayden & Harry

Search/coverage strategy effectiveness	If search is slow or leaves gaps, the demo either times out or fails to find a person who's actually there, regardless of how good detection/delivery are	Implement a simple systematic sweep of a bounded test area and measure time-to-full-coverage in the smaller world before scaling up	Fall back to a simpler systematic sweep pattern over the user-defined search grid if a more adaptive strategy runs out of time	Will & John
Cross-drone location handoff	This is the single link holding the two-drone architecture together - without it, the courier has nowhere to go, and the stretch scenario collapses to a single drone even if both drones work individually	Run two drone namespaces simultaneously early and test whether a point found in one robot's map can be translated into a goal the second robot can actually reach correctly	Use a second drone not included in the package from an external website	Will & John
Simulation performance at full scale	If the full two-robot setup runs too slowly, the team can't reliably test or demonstrate the actual scenario the project is about, regardless of whether the logic is correct	Launch the two-robot configuration with full mapping/navigation/visualisation early, on the weakest laptop on the team, and check it runs at a usable rate	Reduce sensor/visualisation update rates, test primarily on a stronger lab machine, or scale back visualisation for the live demo if needed	Jacob
Drone collision	To ensure a realistic simulation, actions need to be put in place to ensure the drones do not collide especially upon dropping off the first aid kit.	See if preliminary system has collisions or if any path planned has collisions	Run the drones at separate Z values	Jayden

4. Planning and Current Evidence

4.1 Development and Integration Workflow

The repository structure will be based on the ROS package with each part of the program being separated into the respective folder (GUI, path finding, object detection etc). Code merging will be sorted via separate branches on GitHub where applicable with each branch being allocated to a team member/group based on the system they are developing. Task tracking will be allocated to a notion workspace that will feature all the team's tasks in conjunction with an appropriate timeline to give a visual representation of the team's progress and foresight of upcoming tasks. Software interfaces will be initially defined separately by each subsystem with communication between each subsystem being organised once a basic structure/skeleton code has been created, after which connection between each subsystem will be established.

Components should be integrated as frequently as possible, especially when making changes to critical systems such as motion/path finding. This further links to testing where integration is essential when testing a complete part of the system (e.g. testing initial search pattern system with GUI, obstacle avoidance and path finding). Testing should also be performed as frequently as required based on the scale of a change made. Large changes to features or development of features should be tested against the last stable build of the entire program, except in cases where a team member is confident that a new or unstable version of the repo features the bare essential functioning code. All technical decisions should be documented on Notion in conjunction with code comments and commit messages on Github. If any major decisions are to be made that branch out of the scope of a team member's individual component, consolidation with the group is strongly encouraged prior to proceeding with a commit/merge.

4.2 Milestones, Gantt Chart and Immediate Next Steps

The project schedule (Appendix A) spans 18 calendar weeks from Week 0 (20 July) to final submission deadline (20 November). It is structured around three phases and eight milestones aligned to the assessment deadlines. Phase 1 establishes scope, requirements and the development environment. Phase 2 opens with two risk-reduction spikes testing perception feasibility and simulation fidelity, then develops the perception, decision-making and user-interface components in parallel against agreed node interfaces. The first working component is targeted at M3 (16 September) and the first end-to-end prototype at M4 (25 September), ahead of the C3 review. Phase 3 delivers advanced behaviours, full system integration (M6), and a two-week testing and evaluation period preceding the live demonstration.

4.3 Roles and Responsibilities

Activity or responsibility	Lead	Support	Expected contribution or outcome
Perception	Jayden	Harry	Reliable RGB (and thermal) person detection sends confirmed locations to the mission coordinator.
Decision making	John	Will	Search-grid path planning for the scout drone and delivery path planning for the courier drone.
User interface	Jacob		A working mission dashboard for setting the search area,

			confirming detections, and monitoring both drones.
Simulation development	Harry	All members	A forest-like Gazebo world with relevant features (person model, drop zones, obstacles) built on the provided package.
System integration	Jacob	All members	Perception, decision-making, and UI subsystems are connected across both drone namespaces.
Testing and evaluation	Will	John	Early and ongoing testing of individual components and full mission runs against the requirements table.
Repository and software quality	Jacob	All members	Ensure sensible use and management of github to manage code iterations and variations
Stakeholder communication	John	Will	Communicate with stakeholder to align system requirements with stakeholders purpose of use
Project documentation	Harry	Jayden	Document all key decisions made and progress reports
Project management	Will	John	Manage gaant chart and notion to organise tasks effectively.

4.4 Current Technical Progress

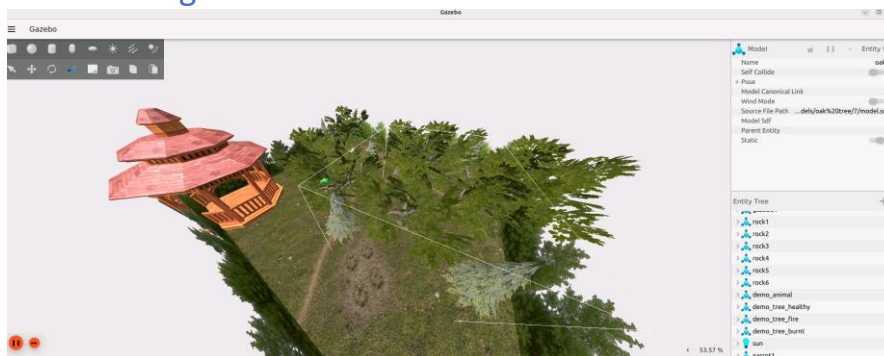


Figure 3- large_demo forest simulation environment in Gazebo

Since receiving the project brief, the team has installed and verified the provided ROS 2 Humble / Ignition Gazebo Fortress environment, confirming the starter package builds and runs correctly.

Beyond reading the documentation, we tested the provided simulation environment hands-on. With SLAM Toolbox enabled, we manually drove the Parrot drone through the large_demo forest world and as the drone explores, we confirmed the occupancy-grid map builds correctly from lidar data (Figure 4). With Nav2 also enabled, we used RViz's goal-pose tool to send navigation goals and confirmed the drone plans a path and drives to the target successfully (Figure 5).

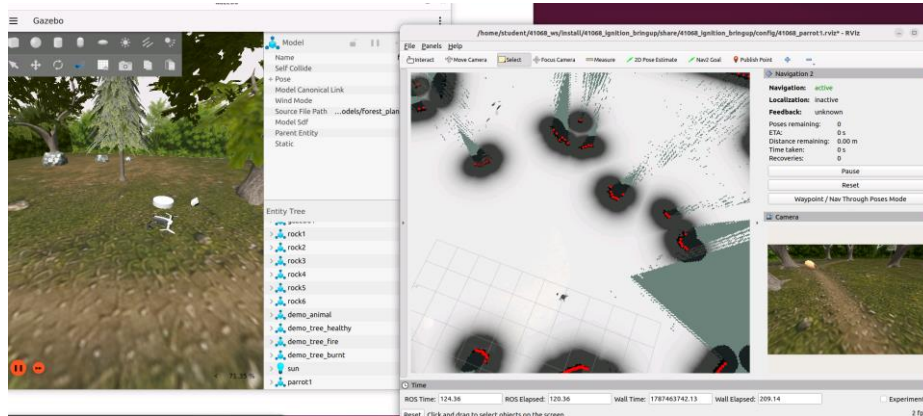


Figure 4- RViz showing the occupancy-grid map being built in real time from lidar data as the Parrot drone is manually driven through the large_demo world.

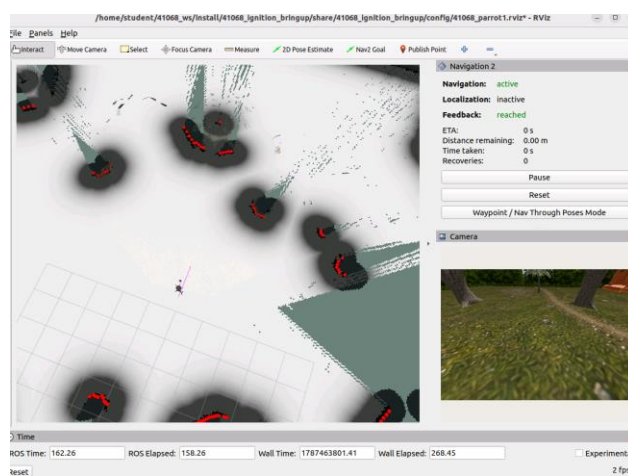


Figure 5- RViz Nav2 panel showing a navigation goal successfully reached ("Feedback: reached"), confirming autonomous path planning and execution.

We also ran the provided basic_autonomy_demo.py example, which combines simple perception (camera brightness) and decision-making (goal selection) to autonomously send goals to Nav2. Terminal output confirms the full goal-sending, acceptance, and completion cycle executes correctly (Figure 7).

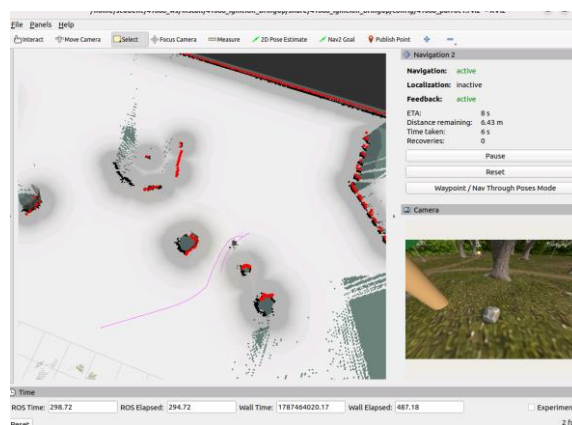


Figure 6- RViz Nav2 showing the drone path during the basic_autonomy_demo.py example.

```

student@labNVXMF8: ~$ ros2 launch 41668_ignition_bringup 41668_autonomy_demo.launch.py robot:=parrot1
[INFO] [launch]: All log files can be found below /home/student/.ros/log/2026-08-23-05-46-25-039632-labNVXMF8-12427
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [basic_autonomy_demo.py-1]: process started with pid [12428]
[parrot1.basic_autonomy_demo]: Autonomy demo started for robot "parrot1". Using map frame "parrot1_map" and base frame "parrot1_base_link".
[parrot1.basic_autonomy_demo]: Waiting for Nav2 NavigateToPose action server...
[parrot1.basic_autonomy_demo]: Brightness=0.50; choosing a farther free-space goal at (14.49, -6.43).
[parrot1.basic_autonomy_demo]: Sending Nav2 goal: (14.49, -6.43) in frame parrot1_map.
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 9.94 m.
[parrot1.basic_autonomy_demo]: Nav2 goal accepted. Waiting for result...
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 8.98 m.
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 5.98 m.
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 3.98 m.
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 1.18 m.
[parrot1.basic_autonomy_demo]: Nav2 goal finished: SUCCEEDED.
[parrot1.basic_autonomy_demo]: Brightness=0.28; choosing a farther free-space goal at (5.79, 8.17).
[parrot1.basic_autonomy_demo]: Sending Nav2 goal: (5.79, 8.17) in frame parrot1_map.
[parrot1.basic_autonomy_demo]: Nav2 goal accepted. Waiting for result...
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 9.98 m.
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 9.31 m.
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 6.23 m.
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 3.46 m.
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 1.12 m.
[parrot1.basic_autonomy_demo]: Nav2 goal finished: SUCCEEDED.
[parrot1.basic_autonomy_demo]: Brightness=0.28; choosing a farther free-space goal at (0.24, 8.34).
[parrot1.basic_autonomy_demo]: Sending Nav2 goal: (0.24, 8.34) in frame parrot1_map.
[parrot1.basic_autonomy_demo]: Nav2 goal accepted. Waiting for result...
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 9.90 m.
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 8.63 m.
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 5.41 m.
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 2.46 m.
[parrot1.basic_autonomy_demo]: Nav2 goal finished: SUCCEEDED.
[parrot1.basic_autonomy_demo]: Brightness=0.28; choosing a farther free-space goal at (2.24, -4.71).
[parrot1.basic_autonomy_demo]: Sending Nav2 goal: (2.24, -4.71) in frame parrot1_map.
[parrot1.basic_autonomy_demo]: Nav2 goal accepted. Waiting for result...
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 0.98 m.
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 11.36 m.
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 9.73 m.
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 6.56 m.
[parrot1.basic_autonomy_demo]: Nav2 feedback: distance remaining 3.39 m.
[parrot1.basic_autonomy_demo]: Nav2 goal finished: SUCCEEDED.
[parrot1.basic_autonomy_demo]: Brightness=0.10; choosing a farther free-space goal at (17.28, -3.65).

```

Figure 7- Terminal output during the `basic_autonomy_demo.py`

Finally, we checked system-level integration using ROS 2's own inspection tools: `ros2 node list` confirms all expected nodes are running together (Figure 8), and `rqt_graph` shows the running nodes connected via ROS topics, confirming the software is wired together as a single connected system rather than isolated pieces (Figure 9).

```

student@labNVXMF8: ~$ ros2 node list
/parrot1/basic_autonomy_demo
/parrot1/bt_navigator
/parrot1/bt_navigator.navigate_through_poses_rclcpp_node
/parrot1/bt_navigator.navigate_to_pose_rclcpp_node
/parrot1/controller_server
/parrot1/gazebo_bridge
/parrot1/global_costmap/global_costmap
/parrot1/lifecycle_manager/navigation
/parrot1/local_costmap/local_costmap
/parrot1/planner_server
/parrot1/robot_localization
/parrot1/robot_state_publisher
/parrot1/rviz2
/parrot1/rviz.navigation_dialog_action_client
/parrot1/slam_toolbox
/parrot1/smooth_server
/parrot1/transform_listener_tmpl_57621ff6d430
/parrot1/transform_listener_tmpl_5847d72eb330
/parrot1/transform_listener_tmpl_5933e8975130
/parrot1/transform_listener_tmpl_5d35ef992c0
/parrot1/transform_listener_tmpl_5e5787642590
/parrot1/transform_listener_tmpl_5f6d7b95d180
/parrot1/transform_listener_tmpl_62eda27ba850
/parrot1/velocity_smoother
/parrot1/waypoint_follower
/ros_gz_bridge_clock
/rqt_gui_py_node_13306
/teleop_twist_keyboard
student@labNVXMF8: ~$

```

Figure 8- Output of `ros2 node list`, confirming the full software stack (bridge, localisation, SLAM, Nav2 servers, and the autonomy node) is running correctly.

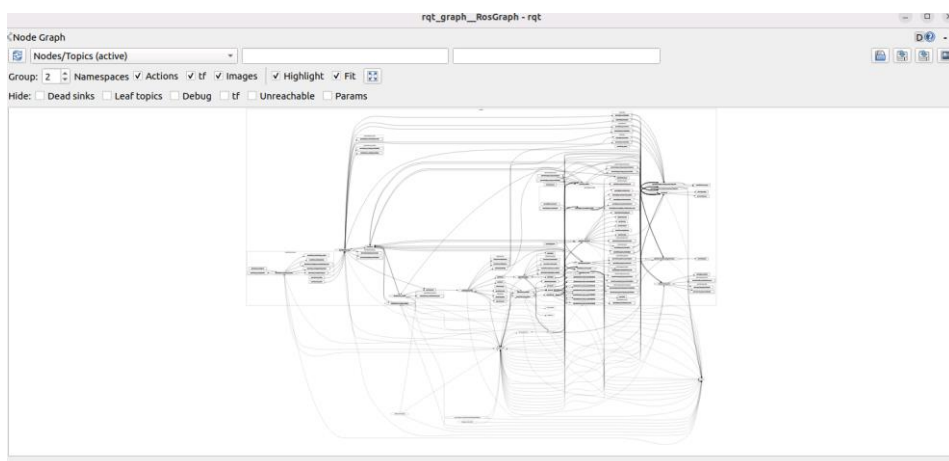


Figure 9- `rqt_graph` showing the running nodes connected via ROS topics, showing that the software stack is wired together as a single connected system rather than isolated pieces

This hands-on testing gave the team a concrete, verified understanding of which components are provided out-of-the-box versus what we will need to build ourselves, directly informing the Component Plan in Section 3.3.

5. Communication

5.1 Team Decision and Action Log

Date	Team members	Decision, issue, or discussion	Summary / outcome
28/7	All Members	Ideation started – created Miro with all ideas for skidsteer and parrot robots	Developed 3 basic ideas each for the skidsteer and parrot.
4/8	All Members	Crazy 8's – Each member created more ideas. Member discussion decided on one.	Our team decided on a safety drone system for our project with double drone capability.
11/8	All Members	Delegated roles according to distribution as spoken to our tutor.	Decided on each individuals contributions and roles within the group Jayden & Harry - Vision and Sensing, Will & John – Movement and Path Planning, Jacob UI and overall integration.
18/8	All Members	Discussed project management systems, general code archetype, communication channels and general coordination.	- Create GitHub Repo for code sharing - Teams for file sharing - Notion website for gannt chart and project management - Instagram for general communication

5.2 Stakeholder Communication Log

Date	Team members	Decision, issue, or discussion	Summary / outcome
4/08/2026	All Members	Asked about feasibility of our first aid project idea. Is person finding a valid idea	Stakeholder is engaged and enthusiastic about the idea and its application
11/08/2026	All Members	Are two drones a valid idea?	Stakeholder communicated that a 2 drone system would be useful and technically feasible given the drones don't work in conjunction simultaneously.
18//08/2026	All Members	Can we add a thermal sensor? Is a randomised environment a valid addition?	Thermal sensor can be equipped for increased accuracy. Random environment is a valid addition, system does not have to be completely random and could instead cycle over a set amount of variations.

5.3 Priority Stakeholder Questions

Question or Uncertainty	Why is this important to clarify?	Response received
Is the two-drone (scout + courier) architecture appropriate for the 12-week timeline, or should we prioritise a fully reliable single-drone MVP before attempting the second drone?	Directly affects whether early development time goes into cross-drone communication or into deepening a single robot's capability first.	Begin using one drone to get a baseline, then increase complexity.
Does it only work when looking for one lost person? Do we need to account for possibility of two?	Directly affects whether early development time goes into cross-drone communication or into deepening a single robot's capability first.	Begin simple to get a baseline, then increase complexity.
How can we break down our team roles? Do we have to do extra considering we have 5 people?	Impacts team dynamic and responsibilities.	2x people on perception, 2x on movement, 1x on UI. Could use 5 th person for advanced task later down the project timeline
How can we actually confirm we found a person? Human based confirmation via UI?	Impacts simulation workflow, perception sub system responsible.	Yes, can use UI to confirm. Look into using thermal sensors maybe.

6. Generative AI

6.1 Generative AI Use Declaration

Tool	Task supported	How the output was used	How the output was checked or verified
Claude	Feedback on our project ideas	Used to highlight potential flaws or areas for improvement	Human research to validate our final conclusions
Claude	Analysis of given package	Helped provide teaching material on what the supplied ROS2 package supported and how to interact with it	Only for learning purposes, any references to code were double checked to be accurate.

6.2 Planned Generative AI Protocol

The team agrees to use AI in the following contexts:

- To help ideate potential solutions to issues and brainstorm different potential approaches to the system

- Generate small coding sections where the user understands all code generated and it is not an extremely large section
 - To help analyse the codebase for potential run-time bugs and fixes and teach the solution
- All AI generated content will be fact checked by primary sources and critically analysed for hallucinations.

7. References

Blecher, D. (2022). For the first time in medical history, an autonomous drone helps save the life of a cardiac arrest patient. In Everdrone. <https://everdrone.com/news/2022/01/04/for-the-first-time-in-medical-history-an-autonomous-drone-helps-save-the-life-of-a-cardiac-arrest-patient/>

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8. Appendix

8.1 Appendix A – Gantt Chart

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